DFA ID

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#include <stdio.h>
#include <string.h>
#include <ctype.h>

char stat_table[3][3][10] = {
    {"stat", "letter", "digit"},
    {"start", "ID", "ID"},"erro"}
    {"ID", "ID", "ID"}
};

int main() {
    char input[20], column_stat[10], next_stat[10], current_stat[10];
    char ch;
    int i, len, row, column, error = 0;

    clrscr(); // Clear the screen
    printf("Enter Identifier: ");
    gets(input); // Avoid using gets() in production code due to security
vulnerabilities
    len = strlen(input);
    printf("OUTPUT:\n");
    strcpy(current_stat, "start");

    for (i = 0; i < len; i++) {
        ch = input[i];
```

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        if (isalpha(ch)) {
            strcpy(column_stat, "letter");
        } else if (isdigit(ch)) {
            strcpy(column_stat, "digit");
        } else {
            strcpy(column_stat, "invalid");
            error++;
        }

        if (error == 0) {
            for (row = 1; row < 3; row++) {
                if (strcmp(stat_table[row][0], current_stat) == 0) {
                    break;
                }
            }

            for (column = 1; column < 2; column++) { // Changed the loop range to
2 since there are only 2 columns now
                if (strcmp(stat_table[0][column], column_stat) == 0) {
                    break;
                }
            }

            strcpy(next_stat, stat_table[row][column]);
            printf("%s %c %s\n", current_stat, ch, next_stat);
        } else {
            printf("%s %c %s\n", current_stat, ch, "invalid");
            printf("ERROR: Invalid Character\n");
```

```
            break;
        }

        strcpy(current_stat, next_stat);
    }

    if (error == 0 && strcmp(current_stat, "ID") == 0) {
        printf("Valid Identifier\n");
    } else {
        printf("Invalid Identifier\n");
    }

    getch(); // Wait for a key press
    return 0;
}
```